

Integrating threat mapping and animal movement data to identify high-risk areas for endangered mobile species

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Keywords

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Abstract

Given the current biodiversity crisis, understanding how animals move across a landscape dotted with different anthropogenic threats and the consequences of those threats for animals is paramount to devising evidence-based conservation interventions. Vultures roam across large areas and are highly exposed to poisoning, which represents a particularly damaging form of wildlife crime. In this study, we introduce a framework for quantifying the exposure to threats and illustrate an example of poisoning risk as a threat in an endangered African vulture species, the Lappet-faced Vulture (*Torgos tracheliotos*). We combined GPS tracking data of 19 individuals collected between 2012 and 2022 with food availability and spatial threat maps of both intentional (poachers directly targeting vultures) and unintentional (farmers aiming to kill carnivores, with vultures being secondarily affected) poisoning across most of Southern Africa. We identified poisoning hotspots in northern Botswana and south-eastern Namibia. These areas were also associated with a high number of vulture mortalities, providing additional support for poisoning risk. Northern Botswana and areas at the border between Botswana and South Africa were characterized by high food availability, potentially amplifying the mortality rate by attracting vultures from surrounding areas. Our results offer valuable insights for regional vulture conservation, together with a methodological framework for quantifying and mapping the spatial exposure to threats for mobile species of conservation concern, enabling improved targeting of conservation actions.

Introduction

Wildlife poisoning is one of the most pervasive threats to biodiversity (Olea *et al.*, 2022), causing cascading effects throughout entire ecosystems (Van Dyk & Pletschke, 2011; Ogada *et al.*, 2016a). Poisoning particularly impacts scavengers, including vultures, which provide important ecosystem services such as waste recycling, disease control and contributing to sustainable development goals (Carucci *et al.*, 2022; Frank & Sudarshan, 2023; Santangeli *et al.*, 2024). As obligate scavengers, vultures rely not only on spatio-temporally

unpredictable sources such as game carcasses but also feed on predictable food sources such as livestock or game hunting carrion. Over the last three decades, vulture species worldwide have faced major population declines, mainly due to poisoning (Ogada, Keesing, & Virani, 2012; Safford *et al.*, 2019). Asian populations have collapsed, presumably due to non-steroidal anti-inflammatory drug diclofenac (Green *et al.*, 2006). A similar trend occurred in Africa, where populations have declined drastically, primarily due to the use of pesticides including organophosphates and carbamates (Ogada, 2014; Ogada *et al.*, 2016a). Habitats with

high levels of poisoning and also high food availability (game and livestock as potential carcasses), might attract vultures, which can result in increased mortality (Santangeli *et al.*, 2016, 2017). Such maladaptive behavior, when an animal chooses a habitat where it performs poorly relative to alternative habitats, can result in that animal being caught in an 'ecological trap' (Robertson & Hutto, 2006).

In Africa, poisoning due to human–wildlife conflict (HWC; hereafter referred to as unintentional poisoning) occurs when farmers suffer livestock losses and subsequently attempt to kill predators in retaliation (e.g. lions, *Panthera leo*; leopards, *Panthera pardus*; and jackals, *Canis* spp.) but indiscriminately kill vultures that consume poisoned bait (Ogada, 2014; Santangeli *et al.*, 2016, 2017). Sentinel poisoning (Ogada, Botha, & Shaw, 2016b; hereafter referred to as intentional poisoning) occurs when poachers intentionally poison vultures in an attempt to curb the birds' sentinel role of circling in the sky above carcasses, which alerts authorities conducting anti-poaching operations focused on elephants and other wildlife (Wittemyer *et al.*, 2014; Nhleko *et al.*, 2022). With sentinel poisoning, poachers typically leave a carcass, sometimes of a very large animal, like an elephant, with poison (Margalida, Ogada, & Botha, 2019). If not detected quickly and decontaminated, such poisoned carcasses have the potential to kill tens or even hundreds of vultures in a few days (Murn & Botha, 2018), as occurred for example in Botswana, where several hundred vultures were poisoned in a single incident in 2013 (Leepile *et al.*, 2020).

Southern Africa, while representing a global priority region for vulture conservation, also faces significant challenges for conserving these species due to the concomitant presence of two types of poisoning, among other threats such as electrocution and collisions caused by power infrastructure, decline of food availability and habitat loss (Botha *et al.*, 2017; Santangeli *et al.*, 2019). The highly mobile nature of vultures further magnifies the scale of these challenges, as these animals move across extensive landscapes (see e.g. Spiegel *et al.*, 2015; Kane *et al.*, 2022) with potentially varying poison risk levels. Understanding how vultures move across and use the landscape relative to the variation in the risk of different poisoning practices represents a first critical step toward addressing the vulture conservation challenge. This knowledge would allow the identification of priority areas where vultures face a disproportionate risk of being killed due to poisoning practices, and design targeted conservation strategies focused on the most critical areas of risk identified.

To date, we lack a unified framework to identify areas with high spatial exposure to poisoning, which can be used for other threats as well. Therefore, in this study, we set out a conceptual and methodological framework for quantifying and mapping the spatial exposure of animals to their threats (Fig. 1). We used GPS-tracked Lappet-faced Vultures (*Torgos tracheliotos*) moving in a landscape where poisoning is prevalent (including intentional and unintentional poisoning) as a model system, and we identified poisoning risk areas for high-priority conservation intervention in Southern Africa,

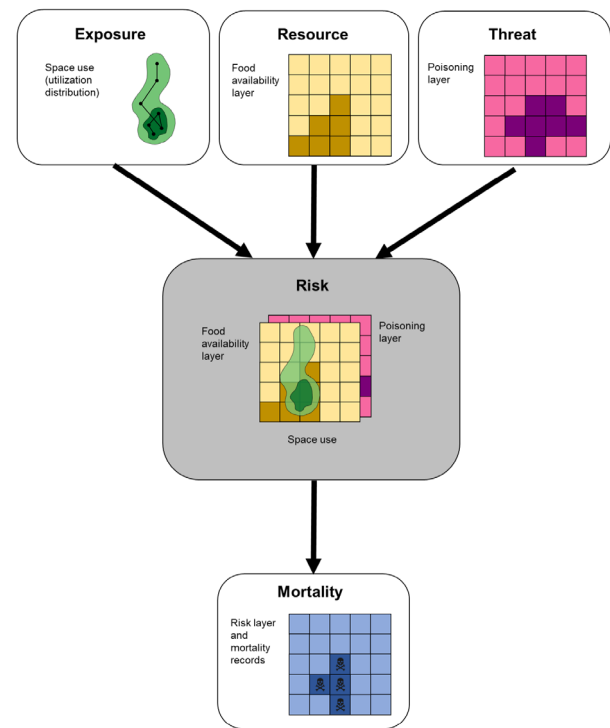


Figure 1 Conceptual and methodological framework. Habitats, where a threat is present (poisoning), that are also rich in resources (food availability) attract vultures, causing high exposure to risk and consequently mortality. Using values of space use (utilization distribution), calculated from GPS trajectories, and spatial layers of food availability and intentional and unintentional poisoning, we map the exposure to poisoning risk (see below). Spatial records of vulture mortality due to poisoning represent the consequences of spatial exposure to poisoning risk.

with implications for other taxa, threats and study areas. Our overarching aim was to quantify the extent of this species' spatial exposure to intentional and unintentional poisoning and to pinpoint areas with high food availability and high poisoning levels that potentially lure vultures into dangerous locations. In addition, we compared spatial exposure to poisoning risk with existing records of vulture mortality to assess the consequences of poisoning risk.

Materials and methods

GPS tracking and data preparation

We captured 20 adult Lappet-faced Vultures using traps baited with ungulate carcasses and noose loops or a gas-propelled cannon net system (WCS NetBlaster™, Wildlife Control Supplies, East and Granby, CT 06026, USA). We equipped vultures with 70 g solar-powered GPS PTT-100 tags (Microwave Telemetry Inc., Maryland, USA). For details on the study species, see supporting information and for details regarding vulture capture and tagging, see Garbett (2018).

Because we sampled GPS positions at different schedules, we thinned movement data to every 2 h, using the ‘thin-TrackTime’ function in the R package ‘move’ (Kranstauber *et al.*, 2023) and kept only positions during the day, between 4:00 am and 6:00 pm UTC (mean $\pm$ SE fixes per day: 7.2 ± 0.01), to match the sampling time of our datasets and thus maximize the number of positions. We additionally removed duplicate timestamps, missing coordinates, outliers and trajectories with less than 100 GPS positions. After data cleaning, the resulting dataset consisted of 19 individuals, 52 annual trajectories according to the calendar year, and 85,752 data points from 2012 to 2022 (mean $\pm$ SE duration of tracking per individual: 654.5 ± 61.5 days).

Utilization distribution

From annual trajectories, we calculated Utilization Distribution (UD) to identify areas individuals occupied during their annual cycle. We applied a dynamic Brownian bridge movement model (dBBMM) using the R package ‘move’ (Kranstauber *et al.*, 2023). The dBBMM in comparison to traditional minimum convex polygons (MCP) and kernel density estimators (KDE), quantifies UD based on an animal’s movement path (rather than individual points), it accounts for spatial uncertainty and temporal autocorrelation and can handle heterogeneous datasets and irregular sampling schedules (Kranstauber *et al.*, 2012). MCP and KDE are used for estimating home range area while Brownian bridge estimators are used for answering the question of where an animal was located during the observation period (Fleming *et al.*, 2015). The dBBMM takes into account changes in behavior along the movement path when calculating Brownian motion variance (an estimate of how regular the movement is), using a moving window with n locations within which the breakpoints of behavioral change are detected. Following Kranstauber *et al.* (2012), we selected the values of window size based on the temporal resolution of the data and our a priori assumption of the time we expect behavioral change. Next to window size, the model requires margin size; that is, the number of ≥ 3 locations on both ends of the sliding window where breakpoints are not detected, and location error of the positions.

We calculated the dynamic Brownian motion variance with a window size of 21 locations (corresponding to 3 days), and a margin of seven locations (corresponding to 1 day) since we expected that during the day when GPS data were recorded, vultures would switch between faster movement (e.g. traveling) and slower movement (e.g. searching for food), and a location error of 50 m (a conservative estimation of location error in our data). We then computed dBBMM using a raster of intentional poisoning (see below) as a baseline with a 10 km grid cell size and a default value of 0.3 (30%) for the extension of the bounding box around the animals’ tracks. From dBBMM, we created a UD raster using the ‘getVolumeUD’ R function and removed the upper 1% of values to get 99% UD. The resulting UD values represent the proportional probabilities of an individual being located in a specific area/raster cell during its annual cycle.

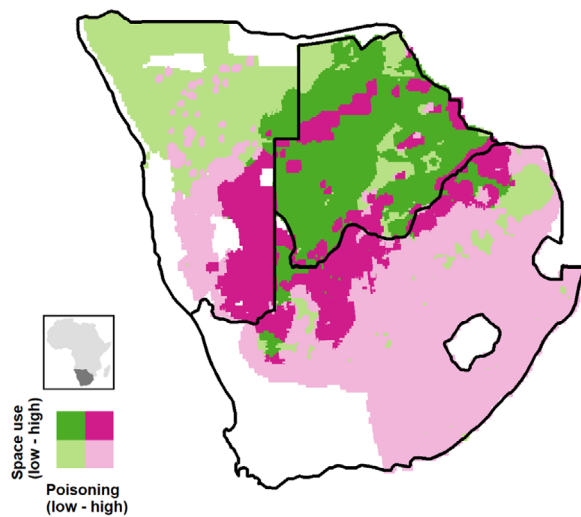
Food availability and poisoning layers

As a measure of food availability, we used total biomass, obtained as a raster layer at a 50 km resolution which was specifically developed for Africa (Hempson, Archibald, & Bond, 2015, 2017). The total biomass includes livestock (cattle, sheep and goats) and herbivore biomass (92 herbivore species) from 1958 to 2013 (median = 1995).

We derived the intentional poisoning layer directly from Santangeli *et al.* (2019). The study produced this layer by identifying the herbivore and carnivore species targeted by poachers for sentinel poisoning, with the use of an integrated poisoning database (The Endangered Wildlife Trust and the Peregrine Fund, 2023). Specifically, the search for intentionally poisoned species in the above database indicated that medium- to large-sized herbivore species, along with carnivores such as lion (*Panthera leo*) and leopard (*Panthera pardus*) have been recorded to be intentionally poisoned, with different magnitudes varying by species. Two steps were implemented to minimize any effect of the potential species detection and reporting bias in the above database. First, error bias was reduced (whereby an intentionally poisoned species is absent from the database) by conservatively selecting all African Cetartiodactyla and Perissodactyla species (as species within these two orders have been targeted) with body mass ≥ 20 kg that were not included in the poisoning database, and assigned them the minimum frequency score of one (Santangeli *et al.*, 2019). This first step yielded a complete list of the targeted species ($n = 72$ in total) and their relative occurrence in the poisoning database (including those not recorded but belonging to targeted groups and thus having a frequency of 1). Second, the relative occurrence of each of the 72 species in the poisoning database was used as a proxy for the species’ likelihood of being intentionally poisoned. At this step, the relative occurrence values were conservatively log (plus two) transformed, so that relative differences among species are reduced, to minimize the bias associated with the frequency of species occurrence in the poisoning database. This conservative species-specific value of poisoning intensity was then combined with the range map of each species, and for each pixel (10 km x 10 km size) of the study region, the mean of the species-specific poisoning intensity values across all species (among the above 72 species) present within that pixel was calculated (for more details, including validation, see Santangeli *et al.*, 2019).

We obtained the unintentional poisoning layer from two empirical and spatially explicit country-wide poison prevalence studies in Namibia (Santangeli *et al.*, 2016, Fig. 4; Craig *et al.*, 2019, Fig. 2) and one in South Africa (Brink *et al.*, 2021, Fig. 3). These studies used specific questioning techniques (with sample sizes of hundreds of in-person survey participants) that permit robust quantification of the prevalence of sensitive behaviors, such as the use of poisons (Nuno & John, 2015). Specifically, we used unintentional poison prevalence raster layers from Namibia’s commercial farmland (Santangeli *et al.*, 2016) relative to the year 2015, and from Namibia’s communal farmland (Craig *et al.*, 2019)

(a) Unintentional poisoning



(b) Intentional poisoning

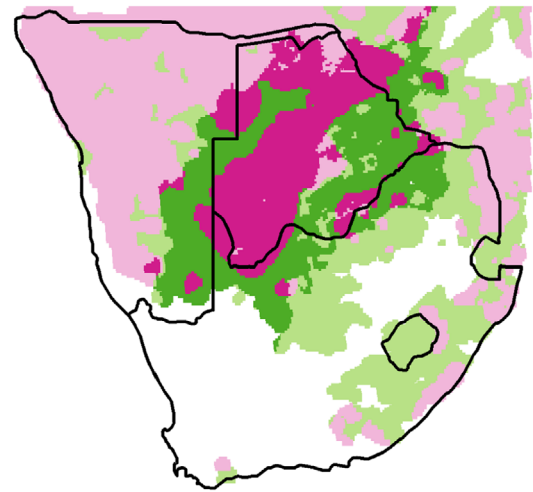


Figure 2 Spatial overlap between space use (utilization distribution – UD) and (a) unintentional poisoning (i.e. poisoning due to human-wildlife conflict) or (b) intentional (sentinel) poisoning. UD represents proportional probabilities of all 52 annual trajectories (19 individuals) jointly occupying a specific area. For breaks in the legend, we used quantiles with probabilities at 0, 0.5 and 1 (in cases when 0.5 quantile was 0, we included the nearest quantile – 0.6 with values >0).

Food availability

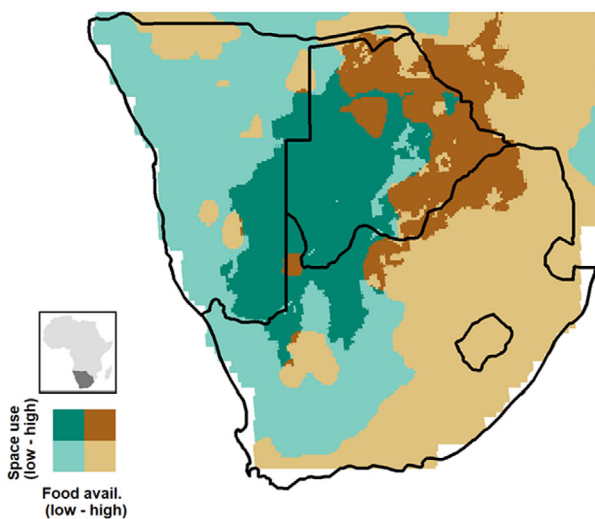


Figure 3 The overlap between space use (utilization distribution – UD) and food availability. UD represents proportional probabilities of all 52 annual trajectories (19 individuals) jointly occupying a specific area. For breaks in the legend, we used quantiles with probabilities at 0, 0.5 and 1 (in cases when 0.5 quantile was 0, we included the nearest quantile – 0.7 with values >0).

relative to the year 2016. Poison prevalence was quantified separately for these two land tenures (commercial and communal farmland) because they exhibit very different land and livestock management practices, which also reflect a large difference in unintentional poison use. Poison use is more

prevalent in commercial than communal areas of Namibia (Craig *et al.*, 2019). For South Africa, we used an unintentional poison prevalence raster layer (Brink *et al.*, 2021) relative to the year 2019 which only covered the commercial freehold farmland of South Africa within the breeding range of any local culture population.

For Botswana, no such studies directly quantifying the extent of poisoning exist. Therefore, we extrapolated poisoning levels based on land tenure using values from Namibia. Specifically, we obtained a land tenure map of Botswana (from Botswana National Atlas 2001, Department of Surveys and Mapping) and considered the following three classes: communal farmland areas, commercial freehold areas and protected areas (including national parks and game reserves where no livestock farming is allowed, and hence no human-wildlife conflict exists). Next, for each of the three land use classes we associated a constant poisoning level within the same land use. We used the poison prevalence for the same land use classes reported from neighboring Namibia. That is, we associated a poisoning level of 1.7% to all communal areas, 20% to all commercial areas and 0% to all protected areas in Botswana. We justified the latter by the fact that since there is no livestock herding, there cannot be human-wildlife conflict (at least of the type we examine) or unintentional poisoning. Since the above poison layers assume that poisoning in Botswana has a similar prevalence as in Namibia, we also produced conservative raster layers for Botswana by considering $\pm 10\%$ from the average poison prevalence, and by doubling (+100%) the poison prevalence for this country as an extreme scenario. We then used the latter rasters, considering poison prevalence uncertainty for Botswana, only for sensitivity analyses to confirm the

robustness of the results against variation in poison prevalence. The final step entailed combining the three unintentional poison layers into a single raster that we used for the analyses.

For the analysis, we reprojected the sentinel intentional poisoning raster, changed the resolution of the food availability raster to 10 km, cropped all rasters to the same extent, and rescaled values of all rasters to a range from 0 to 1. For each UD value, we extracted the value of intentional and unintentional poisoning and food availability as well as different values of unintentional poisoning for sensitivity analysis, using the R package 'raster' (Hijmans *et al.*, 2015). Additionally, we illustrated spatial patterns of overlapping areas of poisoning or food availability (or multiplied values of poisoning and food availability) with proportional probabilities of all individuals occupying these areas. For easier representation, we summed the UD values of all individual annual trajectories per raster cell and then divided each UD value with the number of raster cells so that the values summed to 1. The values thus represent proportional probabilities of all 52 annual trajectories (19 individuals) jointly occupying a specific area. The resulting hotspots represent areas with high levels of poisoning or high food availability that are used by a large number of individuals, intensively by one individual, or combinations of the two. However, in the statistical analysis, we measured the number of vultures occupying an area and individual space use, separately.

Additionally, we tested whether poisoning levels at vulture capture locations were representative samples of Botswana. Capture locations were distributed across Botswana (see Garbett, 2018) and poisoning values at the 50 km radius around locations where vultures were captured varied between 0.01 and 0.5 for intentional and between 0.003 and 0.3 for unintentional poisoning (Fig. S1). These values fall within the mean intentional (0.2) and unintentional (0.04) poisoning in the entire area of Botswana. We thus assume our capture locations represent a relatively random sample.

Statistical analysis

To investigate whether the UD of an individual is higher in areas with high food availability and high intentional and unintentional poisoning than in areas with low food availability and low poisoning, we fit a generalized linear mixed model using 'glmmTMB' R package (Brooks *et al.*, 2022) with beta family and logit link. Because the beta model accepts only values >0 and <1 , we rescaled our variables to a range between 1×10^{-10} and 0.9 (10 decimal places). To avoid bias toward individuals with several years of data, we performed model averaging of 10 generalized linear mixed models with data of one random annual trajectory per individual. From each dataset used in the model, we sampled 10,000 random rows (each row corresponding to one UD value per individual annual trajectory) since model averaging requires models with the same amount of data. We performed model averaging separately for intentional and unintentional poisoning. We ran models with UD as the response variable and individual annual trajectory identity as a random

effect. First, we included an interaction between intentional poisoning and food availability as a predictor, and second, we included an interaction between unintentional poisoning and food availability as a predictor.

To assess if the number of vultures is higher in areas with high levels of intentional and unintentional poisoning and high food availability than in areas with low food availability and low levels of poisoning, we calculated the number of individuals that occupied each raster cell (regardless of UD value) and fit two generalized linear models (GLMs) with the number of vultures as a response variable and quasi-Poisson family. As a predictor in the first model, we included the interaction between intentional poisoning and food availability, and as a predictor in the second model, we included the interaction between unintentional poisoning and food availability. We used the quasi-Poisson family to account for the overdispersion of the Poisson model assessed with a dispersion test using 'AER' R package (Kleiber & Zeileis, 2008).

For sensitivity analysis of unintentional poisoning extrapolated in Botswana, we ran three additional GLMs with quasi-Poisson family where we included the number of vultures as a response variable and unintentional poisoning as a predictor. In each of the three models, the unintentional poisoning values in Botswana were adjusted to -10% , $+10\%$ and $+100\%$ of the country's average value, respectively, while the values for Namibia and South Africa remained unchanged.

Results

Lappet-faced Vultures faced poisoning threats across large parts of their range (Fig. 2) with the largest exposure to unintentional poisoning being in south-eastern Namibia, across north-central and southern Botswana, and the northern part of South Africa (Fig. 2a). The largest exposure to intentional poisoning was in north-eastern, central and south-western Botswana (Fig. 2b). Food availability was larger in eastern part of Southern Africa than western part, with the highest overlap between space use and food availability in North and East Botswana and along the border with Zimbabwe and South Africa (Fig. 3).

By overlapping areas highly used by vultures, areas with high food availability, and high unintentional poisoning (Fig. 4a,c), we revealed high-risk areas in north-central Botswana and across the border between Botswana and South Africa. Similarly, by overlapping areas intensively used by vultures, high food availability and high intentional poisoning (Fig. 4b) we identified a hotspot area across the larger part of northern Botswana and smaller areas along the border between Botswana and South Africa. This map can also help us identify areas with high poisoning and food availability that are not necessarily used by our study species but can represent a risk for other vultures tagged in different areas (see areas marked with dark green in Fig. 4).

The above patterns were further tested quantitatively by modeling space use in relation to poisoning and food availability (Fig. 5a,b). Vultures on the individual level spent a

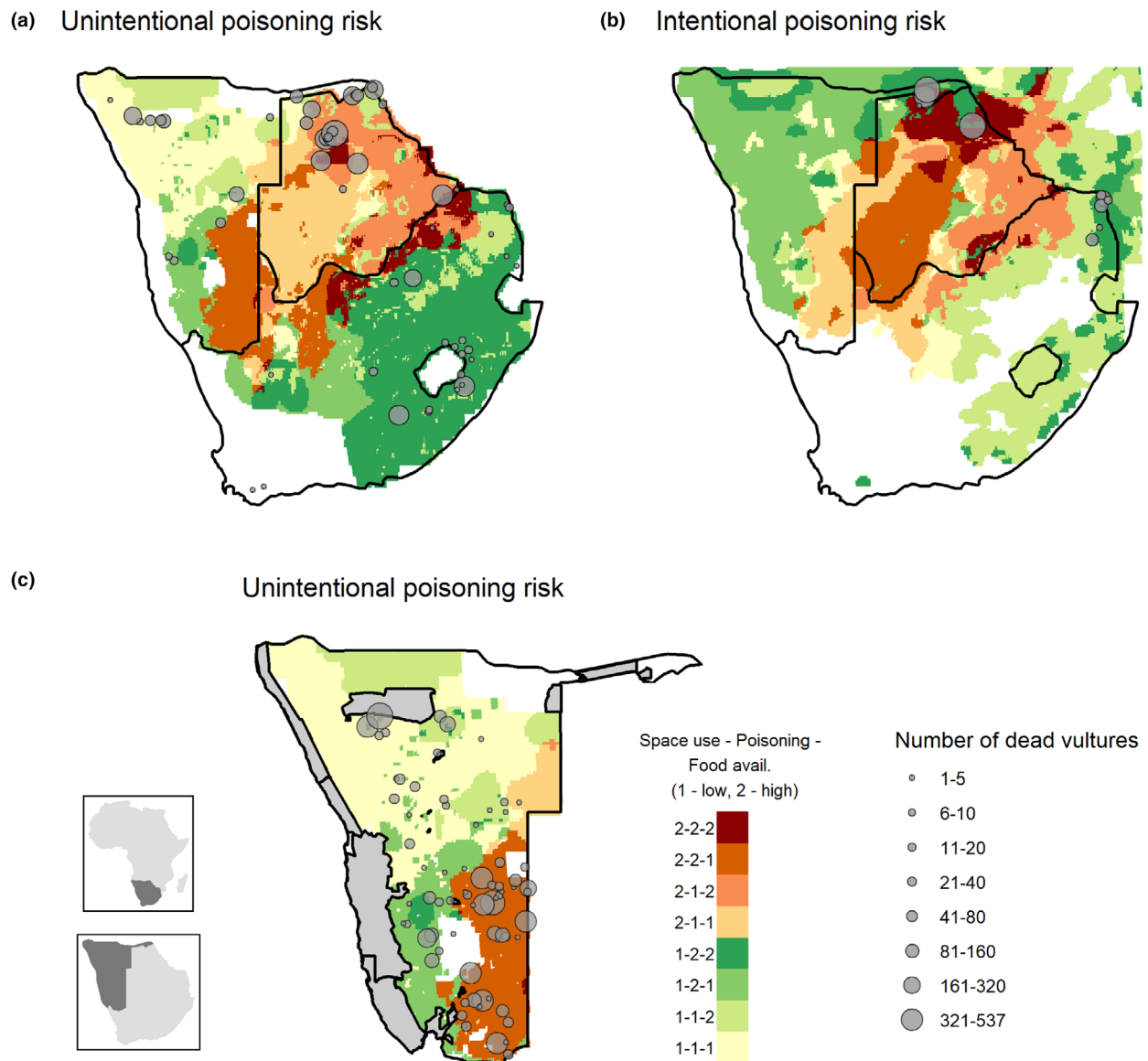


Figure 4 Overlaid space use, poisoning and food availability, and the reported number of vulture mortalities attributed to intentional or unintentional poisoning. (a) Unintentional poisoning data from the African Wildlife Poisoning Database (The Endangered Wildlife Trust and the Peregrine Fund, 2023) of any vulture species for 2000–2023 in Southern Africa ($n = 43$ incidences, range = 1–103, mean = 18.0 vulture mortalities per incident). (b) Reported mortalities from the African Wildlife Poisoning Database (The Endangered Wildlife Trust and the Peregrine Fund, 2023) of any vulture species due to intentional poisoning for 2000–2023 in Southern Africa ($n = 12$ incidences, range = 1–537, mean = 157.9 vulture mortalities per incident). (c) Vulture mortalities of any vulture species due to unintentional poisoning across the commercial farmland of Namibia between 2000 and 2015 ($n = 83$, range = 1–80, mean = 11.0; for details see Santangeli *et al.*, 2017). Light gray areas on the map of Namibia are national parks. For breaks in the legend, we used quantiles with probabilities at 0, 0.5 and 1 (in cases when 0.5 quantile was 0, we included the nearest quantile – 0.6 with values >0).

similar amount of time in areas with low and high levels of unintentional and intentional poisoning as well as food availability (Fig. 5a,b, Tables S1 and S2). Note that Figure 5a,b represents one random model out of 10 models included in the model averaging analysis.

In addition, we used the number of vultures occupying an area instead of the UD to test the same patterns on the

population level (Fig. 5c,d). The number of vultures was significantly higher in areas with high levels of unintentional poisoning under low food availability (Fig. 5c, Table S3), and in areas with high intentional poisoning under high food availability (Fig. 5d, Table S4). We performed the sensitivity to variation in unintentional poisoning in Botswana and the results remained consistent (Tables S5–S7). Note that models

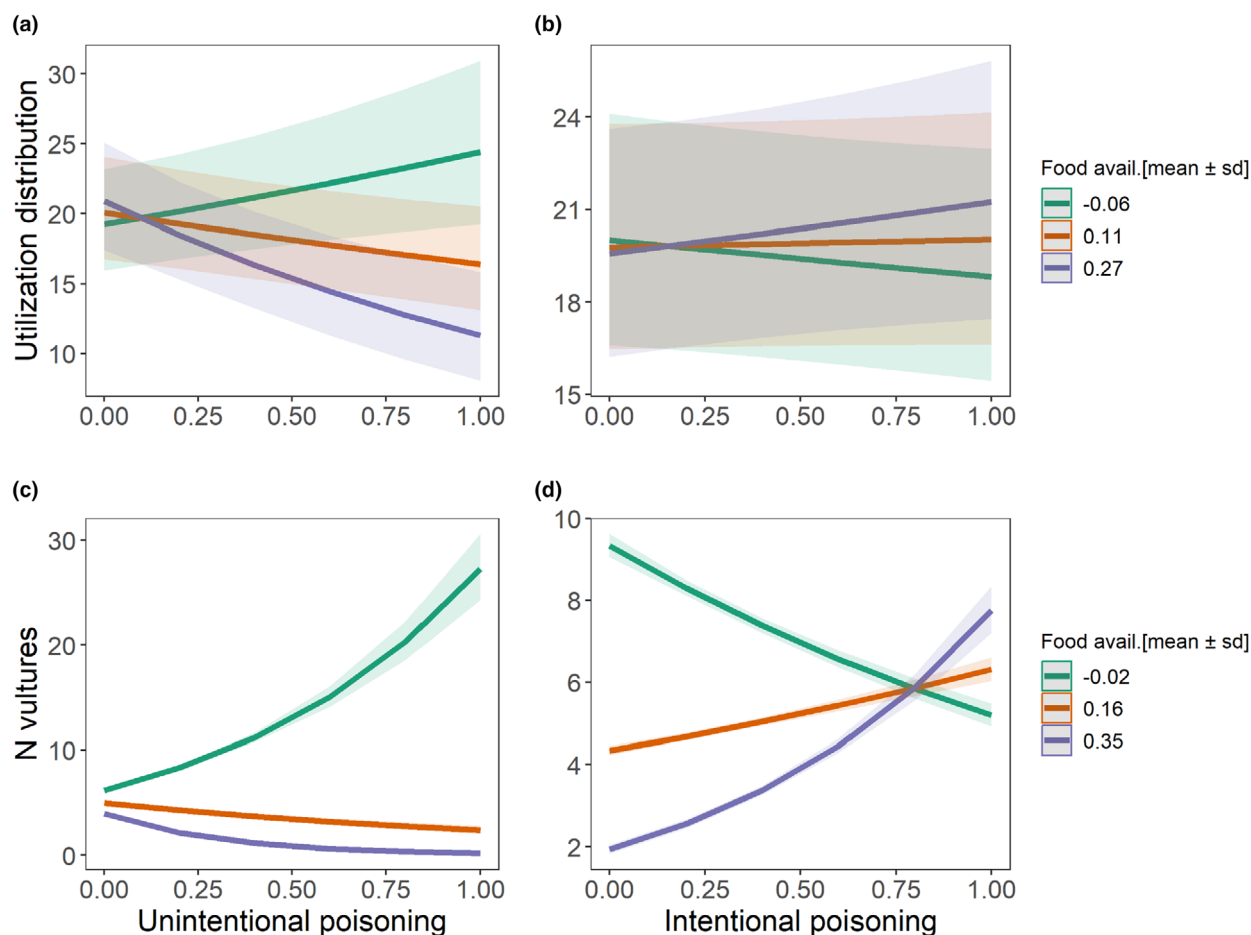


Figure 5 (a and b) Representative generalized linear mixed model with a beta distribution and logit link. We ran models with utilization distributions (UD); that is, the proportional probability of occupying an area as a response variable, interaction between unintentional (a) or intentional (b) poisoning and food availability as a predictor, and individual annual trajectory as a random effect. (c and d) Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between unintentional (c) or intentional (d) poisoning and food availability as a predictor. Colors show representative values of variable food availability from lower limit of SD – green, mean value – orange, and upper limit of SD – purple. Shaded areas show 95% confidence intervals.

testing space use at the population level show clearer results with lower variation than models testing space use at the individual level (Fig. 5), indicating that the hotspots (Fig. 2) occurred mainly due to the high number of individuals occupying these areas.

Northern Botswana, an area of high vulture exposure to poisoning risk and high food availability, was associated with several reported cases of vulture mortalities due to intentional (Fig. 4a) but also unintentional (Fig. 4b) poisoning (reported mortalities were obtained from the African Wildlife Poisoning Database – The Endangered Wildlife Trust and the Peregrine Fund 2023, which includes cases of different vulture species over 2000–2023 in Southern Africa). This pattern of vulture mortalities represents the consequences of the poisoning risk. Another area characterized by a large number of vulture mortalities collected between 2000 and 2015 (mainly Lappet-faced and White-backed, *Gyps africanus*, Vultures; Santangeli *et al.*, 2017) was south-eastern

Namibia (Fig. 4c). This area was linked to high unintentional poisoning but low food availability.

Discussion

In this study, we introduced a framework for quantifying the exposure of endangered and highly mobile animals to threats and illustrated an example of poisoning risk in Lappet-faced Vultures tagged in Botswana. We showed that individuals were exposed to high levels of intentional and unintentional poisoning across a large part of their range. Northern Botswana represents a high-risk area for vultures since intentional and unintentional poisoning as well as food availability were high, potentially amplifying mortality. Indeed, this area was also associated with a high number of vulture mortality records due to intentional and unintentional poisoning, providing additional support for deeming it a high-poisoning risk area. South-eastern Namibia is another high-risk area

associated with high unintentional poisoning and a large number of vulture mortalities. We therefore suggest that conservation practitioners focus on these hotspots to combat the poisoning problem. However, the vulture mortality datasets can have reporting bias so areas identified with high poisoning risk and no or little mortality records mainly covered by commercial farmland, (e.g. the border between Botswana and South Africa (see Shaw *et al.*, 2024)), require special attention.

Northern Botswana is partially covered by reserves, wildlife management areas and commercial farmland with livestock, where both, HWC and poaching prevail, posing a risk of intentional and unintentional poisoning (Garbett, 2018). The area is also associated with high food availability, due to abundant game and livestock, making it attractive to vultures. This potentially represents an ecological trap that may be killing Lappet-faced Vultures from across southern Africa. The same hotspot might represent a risk also for other vulture species, such as the White-backed Vulture from different populations in Southern Africa (Kane *et al.*, 2022). Indeed, a study in north-central Botswana showed a 54% decline in nesting numbers of White-backed Vultures over the last 10 years which was linked to poisoning (Leepile *et al.*, 2020). In addition, population declines of several raptor species occurred in northern Botswana, due to the conversion of natural habitats to farmland (Shaw *et al.*, 2024). Although our tracked Lappet-faced Vultures were trapped and tagged in Botswana, they spent a large proportion of time in south-eastern Namibia. This area was identified with low food availability, but small-stock farming is common and poisoning due to HWC is rampant (Schumann, Walls, & Harley, 2012; Santangeli *et al.*, 2016). Arid conditions triggering small stock mortality which in turn provides small carcasses likely attract this species to this area (Mundy *et al.*, 1992; Spiegel, Getz, & Nathan, 2013), representing a potential ecological trap. Furthermore, potential misperceptions about vultures killing livestock, as has been shown in Europe and South America, can magnify the HWC in these areas (Lambertucci *et al.*, 2021).

The identified hotspots depend on the spatial occurrence of the study species, location and extent of tagging, as well as the ecological requirements of the species. Indeed, age, habitat preference and breeding are expected to affect space use and thus the identified high poisoning risk areas. For example, we characterized KwaZulu-Natal (KZN) as an area with mostly high levels of unintentional poisoning but because Lappet-faced Vultures in our study did not occupy this area, we did not identify it as a hotspot. Another study, however, showed that White-backed Vultures in KZN have low survival due to poisoning (Monadjem *et al.*, 2018). In this study, we indeed show that several vulture mortalities due to unintentional poisoning in KZN and Lesotho have been reported and that the entire eastern part of South Africa represents a potential poisoning risk for vultures utilizing this area since it was associated with high unintentional poisoning and high food availability.

While we aimed to minimize potential biases, our study is not without limitations. First, since data on unintentional

poisoning were not available for Botswana, we extrapolated it based on the best knowledge available. The sensitivity analysis confirmed that the results are robust to variations in unintentional poisoning in Botswana. Nevertheless, this extrapolation should be borne in mind when interpreting the results. Second, the food availability layer includes livestock and wild herbivore species, but we did not account for other sources of food availability, such as butcheries and vulture restaurants, which are expected to affect vulture movements. However, given the scale of our data analysis, we believe that these smaller food sources, in comparison to livestock farms and parks with wildlife, are negligible. Third, identified hotspots largely depend on the space use of the study animal, meaning that the number of individuals, populations and species included in the analysis affects the results. Although our results are relevant for the population of Lappet-faced vultures breeding in Botswana, we suggest that future studies focus on multiple vulture populations and species, combining tracking data from several datasets to identify other potential high-risk hotspots. Additionally, future studies should aim to model space use in relation to different threats, including poisoning, collision and electrocution.

Mortality, especially of adult vultures, can have a profound impact on the persistence of populations since vultures are long-lived, begin breeding only at around 5 years or even 10 years of age, and do not breed every year (Mundy *et al.*, 1992; Sanz-Aguilar *et al.*, 2017; Margalida *et al.*, 2020). Conservation efforts should aim to protect vulture species across their entire life cycle, within and outside protected areas, and across international borders (Santangeli *et al.*, 2018). Mitigating the threats of intentional and unintentional poisoning requires both (1) resolving human–wildlife conflicts, for example, via financial compensation, increasing awareness, diversionary feeding, aversive conditioning and the use of technology (Swan *et al.*, 2017; Brink *et al.*, 2021; Acácio *et al.*, 2023) and (2) addressing the much more subtle sentinel poisoning, wherein poaching is the main driver. Combating poaching in Africa poses a serious contemporary challenge requiring effective patrolling of large areas by rangers with the help of dogs trained to locate poachers and technology, such as camera traps, microphones or GPS tracking of vultures. The use of GPS tracking on vultures allows for monitoring carcasses and potential poaching (Mateo-Tomás *et al.*, 2023). Developing new technologies, such as real-time tags (O'Donoghue & Rutz, 2016) and remote-controlled aircraft (Mulero-Pázmány *et al.*, 2014), would additionally improve poaching detection methods by serving as alert systems for local authorities.

To this end, the Multi-species Action Plan to Conserve African-Eurasian Vultures (Vulture MsAP) under the Convention on the Conservation of Migratory Species of Wild Animals (CMS) (Botha *et al.*, 2017) aims to reverse the population decline of obligate scavenger species of Old World vultures and represents a great opportunity for effective vulture conservation in Africa. Conservation efforts in Africa, however, remain underfinanced, which greatly impairs the effectiveness of protected areas and the successful prevention of vulture poisoning and other biodiversity threats. The

framework targeting increasing financing for biodiversity conservation within the newly ratified program of COP15 (15th Conference of Parties of the UN Convention on Biological Diversity) could greatly help if donors channel financial resources in the right direction in Africa (Obura, 2023). To reverse biodiversity loss in the Global South, we must not only increase the percentage of protected areas but also secure sufficient financial resources to ensure the long-term efficiency and success of conservation projects (Obura, 2023; Botha *et al.*, 2024).

Our study advances the integration of movement ecology and spatial threat mapping, both applied to quantify the spatial risk to wildlife across large regions. This allowed us to identify areas of high concern where conservationists could focus their efforts to achieve the greatest benefit. We reveal high poisoning risk hotspots within northern Botswana and south-eastern Namibia which are potentially affecting populations of threatened vulture species from other countries. These are, to our knowledge, the first confirmed cases, albeit with correlational evidence, of a potential ecological trap affecting vulture species to such a large extent. While the findings of this study are highly worrying, they also represent a clarion call to address the drivers of intentional and unintentional poisoning. Given the drivers and scale of the poisoning threat, leveraging the cooperation between conservation scientists and practitioners from various countries and regions is paramount. On a practical level, all measures to promote a more harmonious coexistence of livestock farmers and wildlife in the region are needed, including compensation schemes, livestock protection measures and educational programs. These initiatives would help reduce unintentional vulture poisoning. Additionally, programs supporting alternative livelihoods for local communities are crucial in minimizing the risk of reverting to poaching, a significant cause of intentional vulture poisoning. Establishing and managing vulture safe zones, designed after the successful examples in Asia, can provide comprehensive protection for key vulture habitats and encourage the adoption of sustainable practices.

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Authors contributions

AS and TC developed the idea and designed the methodology. AA, RG, GM, RR, MS, AB and GT collected movement data. AS and RT provided poisoning layers. FB and GH provided the biomass layer. TC and AS performed the data analysis and wrote the first version of the manuscript. All authors critically revised the manuscript, contributed to the writing and approved the final version of the manuscript. The study is a collaboration between organizations from several countries including Botswana, Namibia and South Africa which is also the spatial extent of our study area. All authors took part in developing this study and contributed their diverse expertise on the topic.

Conflict of interest

The authors declare no conflicts of interest, financial or otherwise.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability statement

Datasets and R code are available via figshare (DOI: [10.6084/m9.figshare.24221470](https://doi.org/10.6084/m9.figshare.24221470)). For additional information and requests, contact the corresponding author.

References

- Acácio, M., Anglister, N., Vaadia, G., Harel, R., Nathan, R., Hatzofe, O. & Spiegel, O. (2023). A lifetime track of a griffon vulture: The moving story of Rehovot (Y64). *Ecology* **104**, e3985.
- Botha, A.J., Andevski, J., Bowden, C.G.R., Gudka, M., Safford, R.J., Tavares, J. & Williams, N.P. (2017). Multi-species action plan to conserve African-Eurasian vultures. *CMS Raptors MOU Technical Publication*. **5**, 1–164.
- Botha, A.J., Doherty, J.E., Weston, J., Andevski, J., Safford, R.J., Tavares, J.P., Bowden, C.G.R., Matsvimbo, F., Sebele, L., AL-hasani, I.K. & Gallo-Orsi, U. (2024). Report on the mid-term implementation review of the multi-species action

- plan to conserve African- Eurasian vultures (vulture MsAP). CMS Secretariat.
- Brink, C.W., Thomson, R.L., Amar, A., Girardello, M. & Santangeli, A. (2021). Prevalence and drivers of poison use by south African commercial farmers and perceptions of alternative livestock protection measures. *Ambio*, **50**, 1211–1221.
- Brooks, M., Bolker, B., Kristensen, K. & Maechler, M. (2022). Package 'glmmTMB'. R Package Version, 1.
- Carucci, T., Whitehouse-Tedd, K., Yarnell, R.W., Collins, A., Fitzpatrick, F., Botha, A. & Santangeli, A. (2022). Ecosystem services and disservices associated with vultures: A systematic review and evidence assessment. *Ecosyst. Serv.* **56**, 101447.
- Craig, C.A., Thomson, R.L., Girardello, M. & Santangeli, A. (2019). The drivers and extent of poison use by Namibia's communal farmers: Implications for averting the African vulture crisis. *Ambio*, **48**, 913–922.
- Fleming, C.H., Fagan, W.F., Mueller, T., Olson, K.A., Leimgruber, P. & Calabrese, J.M. (2015). Rigorous home range estimation with movement data: A new autocorrelated kernel density estimator. *Ecology*, **96**, 1182–1188.
- Frank, E. & Sudarshan, A. (2023). The social costs of keystone species collapse: evidence from the decline of vultures in India. University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2022-165 <https://doi.org/10.2139/ssrn.4318579>
- Garbett, R.A. (2018). Conservation of raptors and vultures in Botswana: with a focus on lappet-faced vultures *Torgos tracheliotos*. PhD thesis, University of Cape Town, South Africa.
- Green, R.E., Taggart, M.A., Das, D., Pain, D.J., Sashi Kumar, C., Cunningham, A.A. & Cuthbert, R. (2006). Collapse of Asian vulture populations: Risk of mortality from residues of the veterinary drug diclofenac in carcasses of treated cattle. *J. Appl. Ecol.* **43**, 949–956.
- Hempson, G.P., Archibald, S. & Bond, W.J. (2015). A continent-wide assessment of the form and intensity of large mammal herbivory in Africa. *Science*, **350**, 1056–1061.
- Hempson, G.P., Archibald, S. & Bond, W.J. (2017). The consequences of replacing wildlife with livestock in Africa. *Sci. Rep.* **7**, 17196.
- Hijmans, R.J., Van Etten, J., Cheng, J., Mattiuzzi, M., Sumner, M., Greenberg, J.A., Lamigueiro, O.P., Bevan, A., Racine, E.B., Shortridge, A. & Hijmans, M.R.J. (2015). Package 'raster'. R package, 734, 473.
- Kane, A., Monadjem, A., Aschenborn, H.O., Bildstein, K., Botha, A., Bracebridge, C., Buechley, E.R., Buij, R., Davies, J.P., Diekmann, M. & Downs, C.T. (2022). Understanding continent-wide variation in vulture ranging behavior to assess feasibility of vulture safe zones in Africa: Challenges and possibilities. *Biol. Conserv.* **268**, 109516.
- Kleiber, C. & Zeileis, A. (2008). *Applied econometrics with R*. New York: Springer-Verlag.
- Kranstauber, B., Kays, R., LaPoint, S.D., Wikelski, M. & Safi, K. (2012). A dynamic Brownian bridge movement model to estimate utilization distributions for heterogeneous animal movement. *J. Anim. Ecol.* **81**, 738–746.
- Kranstauber, B., Smolla, M., Scharf, A.K. & Kranstauber, M.B. (2023). Package 'move'.
- Lambertucci, S.A., Margalida, A., Speziale, K.L., Amar, A., Ballejo, F., Bildstein, K.L., Blanco, G., Botha, A.J., Bowden, C.G.R., Cortés-Avizanda, A., Duriez, O., Green, R.E., Hiraldo, F., Ogada, D., Plaza, P., Sánchez-Zapata, J.A., Santangeli, A., Selva, N., Spiegel, O. & Donazar, J.A. (2021). Presumed killers? Vultures, stakeholders, misperceptions, and fake news. *Conserv. Sci. Pract.* **3**, e415.
- Leepile, L.B., Maude, G., Hancock, P., Reading, R.P., Bridges, B., Hartley, R. & Amar, A. (2020). Changes in nesting numbers and breeding success of African white-backed vulture *Gyps africanus* in north-central Botswana. *Bird Conserv. Int.* **30**, 456–473.
- Margalida, A., Ogada, D. & Botha, A. (2019). Protect African vultures from poison. *Science*, **365**, 1089–1090.
- Margalida, A., Jiménez, J., Martínez, J.M., Sesé, J.A., García-Ferré, D., Llamas, A., Razin, M., Colomer, M. & Arroyo, B. (2020). An assessment of population size and demographic drivers of the bearded vulture using integrated population models. *Ecol. Monogr.* **90**, e01414.
- Mateo-Tomás, P., Rodríguez-Pérez, J., Fernández-García, M., García, E.J., Valente e Santos, J.P., Gutiérrez, I., Olea, P.P., Rodríguez-Moreno, B. & López-Bao, J.V. (2023). Wildlife as sentinels of compliance with law: An example with GPS-tagged scavengers and sanitary regulations. *J. Appl. Ecol.* **60**, 2188–2198.
- Monadjem, A., Kane, A., Botha, A., Kelly, C. & Murn, C. (2018). Spatially explicit poisoning risk affects survival rates of an obligate scavenger. *Sci. Rep.* **8**, 4364.
- Mulero-Pázmány, M., Stolper, R., Van Essen, L.D., Negro, J.J. & Sassen, T. (2014). Remotely piloted aircraft systems as a rhinoceros anti-poaching tool in Africa. *PLoS One*, **9**, e83873.
- Mundy, P., Butchart, D., Ledger, J. & Piper, S. (1992). *The vultures of Africa*, Vol. **671**. London: Academic Press.
- Murn, C. & Botha, A. (2018). A clear and present danger: Impacts of poisoning on a vulture population and the effect of poison response activities. *Oryx*, **52**, 552–558.
- Nhleko, Z.N., Ahrens, R., Ferreira, S.M. & McCleery, R.A. (2022). Poaching is directly and indirectly driving the decline of South Africa's large population of white rhinos. *Anim. Conserv.* **25**, 151–163.
- Nuno, A. & John, F.A.S. (2015). How to ask sensitive questions in conservation: A review of specialized questioning techniques. *Biol. Conserv.* **189**, 5–15.
- Obura, D. (2023). The Kunming-Montreal global biodiversity framework: Business as usual or a turning point? *One Earth*, **6**, 77–80.
- O'Donoghue, P. & Rutz, C. (2016). Real-time anti-poaching tags could help prevent imminent species extinctions. *J. Appl. Ecol.* **53**, 5–10.

- Ogada, D.L. (2014). The power of poison: Pesticide poisoning of Africa's wildlife. *Ann. N. Y. Acad. Sci.* **1322**, 1–20.
- Ogada, D.L., Keesing, F. & Virani, M.Z. (2012). Dropping dead: Causes and consequences of vulture population declines worldwide. *Ann. N. Y. Acad. Sci.* **1249**, 57–71.
- Ogada, D., Shaw, P., Beyers, R.L., Buij, R., Murn, C., Thiollay, J.M., Beale, C.M., Holdo, R.M., Pomeroy, D., Baker, N. & Krüger, S.C. (2016a). Another continental vulture crisis: Africa's vultures collapsing toward extinction. *Conserv. Lett.* **9**, 89–97.
- Ogada, D., Botha, A. & Shaw, P. (2016b). Ivory poachers and poison: Drivers of Africa's declining vulture populations. *Oryx* **50**, 593–596.
- Olea, P.P., Fernández-García, M., López-Bao, J.V., Viñuela, J., e Santos, J.P.V., Rodríguez-Pérez, J., Sotelo, L., Cortizo, C., Sazatornil, V., Bosch, A.P. & Gutiérrez, I. (2022). Unraveling the real magnitude of illegal wildlife poisoning to halt cryptic biodiversity loss. *Biol. Conserv.* **273**, 109702.
- Robertson, B.A. & Hutto, R.L. (2006). A framework for understanding ecological traps and an evaluation of existing evidence. *Ecology* **87**, 1075–1085.
- Safford, R., Andevski, J., Botha, A., Bowden, C.G., Crockford, N., Garbett, R., Margalida, A., Ramírez, I., Shobrak, M., Tavares, J. & Williams, N.P. (2019). Vulture conservation: The case for urgent action. *Bird Conserv. Int.* **29**, 1–9.
- Santangeli, A., Arkumarev, V., Rust, N. & Girardello, M. (2016). Understanding, quantifying and mapping the use of poison by commercial farmers in Namibia—implications for scavengers' conservation and ecosystem health. *Biol. Conserv.* **204**, 205–211.
- Santangeli, A., Arkumarev, V., Komen, L., Bridgeford, P. & Kolberg, H. (2017). Unearthing poison use and consequent anecdotal vulture mortalities in Namibia's commercial farmland—implications for conservation. *Ostrich* **88**, 147–154.
- Santangeli, A., Spiegel, O., Bridgeford, P. & Girardello, M. (2018). Synergistic effect of land-use and vegetation greenness on vulture nestling body condition in arid ecosystems. *Sci. Rep.* **8**, 1–11.
- Santangeli, A., Girardello, M., Buechley, E., Botha, A., Minin, E.D. & Moilanen, A. (2019). Priority areas for conservation of Old World vultures. *Conserv. Biol.* **33**, 1056–1065.
- Santangeli, A., Lambertucci, S.A., Margalida, A., Carucci, T., Botha, A., Whitehouse-tedd, K. & Cancellario, T. (2024). The global contribution of vultures towards ecosystem services and sustainability—an experts' perspective. *iScience* **27**, 10992.
- Sanz-Aguilar, A., Cortés-Avizanda, A., Serrano, D., Blanco, G., Ceballos, O., Grande, J.M., Tella, J.L. & Donazar, J.A. (2017). Sex-and age-dependent patterns of survival and breeding success in a long-lived endangered avian scavenger. *Sci. Rep.* **7**, 40204.
- Schumann, B., Walls, J.L. & Harley, V. (2012). Attitudes towards carnivores: The views of emerging commercial farmers in Namibia. *Oryx* **46**, 604–613.
- Shaw, P., Ogada, D., Dunn, L., Buij, R., Amar, A., Garbett, R., Herremans, M., Virani, M.Z., Kendall, C.J., Croes, B.M. & Odino, M. (2024). African savanna raptors show evidence of widespread population collapse and a growing dependence on protected areas. *Nat. Ecol. Evol.* **8**, 45–56.
- Spiegel, O., Getz, W.M. & Nathan, R. (2013). Factors influencing foraging search efficiency: Why do scarce lappet-faced vultures outperform ubiquitous white-backed vultures? *Am. Nat.* **181**, E102–E115.
- Spiegel, O., Harel, R., Centeno-Cuadros, A., Hatzofe, O., Getz, W.M. & Nathan, R. (2015). Moving beyond curve fitting: Using complementary data to assess alternative explanations for long movements of three vulture species. *Am. Nat.* **185**, E44–E54.
- Swan, G.J., Redpath, S.M., Bearhop, S. & McDonald, R.A. (2017). Ecology of problem individuals and the efficacy of selective wildlife management. *Trends Ecol. Evol.* **32**, 518–530.
- The Endangered Wildlife Trust and the Peregrine Fund. (2023). The African Wildlife Poisoning Database. Downloaded from www.awpd.cloud on 2023-02-01.
- Van Dyk, J.S. & Pletschke, B. (2011). Review on the use of enzymes for the detection of organochlorine, organophosphate and carbamate pesticides in the environment. *Chemosphere* **82**, 291–307.
- Wittemyer, G., Northrup, J.M., Blanc, J., Douglas-Hamilton, I., Omondi, P. & Burnham, K.P. (2014). Illegal killing for ivory drives global decline in African elephants. *Proc. Natl. Acad. Sci.* **111**, 13117–13121.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix S1. Study species and landscape.

Figure S1. Distribution of (a) unintentional and (b) intentional poisoning values within a 50 km radius around capturing locations of vultures in Botswana. The dashed line represents the mean poisoning value of the entire country.

Table S1. Model averaging of 10 generalized linear mixed models with data of one random annual trajectory per individual. We fit the model with a beta distribution and logit link and included utilisation distribution (UD) (i.e., the proportional probability of occupying an area) as a response variable, interaction between unintentional poisoning and food availability as a predictor, and individual annual trajectory as a random effect

Table S2. Model averaging of 10 generalized linear mixed models with data of one random annual trajectory per individual. We fit the model with a beta distribution and logit link and included utilisation distribution (UD) (i.e., the proportional probability of occupying an area) as a response variable, interaction between intentional poisoning and food availability as a predictor, and individual annual trajectory as a random effect

Table S3. Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between unintentional poisoning and food availability as a predictor

Table S4. Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between intentional poisoning and food availability as a predictor

Table S5. Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between unintentional poisoning (with -10% from the average poisoning prevalence in Botswana)

and food availability as a predictor

Table S6. Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between unintentional poisoning (with +10% from the average poisoning prevalence in Botswana) and food availability as a predictor

Table S7. Generalized linear model with a quasi-Poisson distribution and the number of vultures as a response variable, the interaction between unintentional poisoning (with +100% from the average poisoning prevalence in Botswana) and food availability as a predictor